

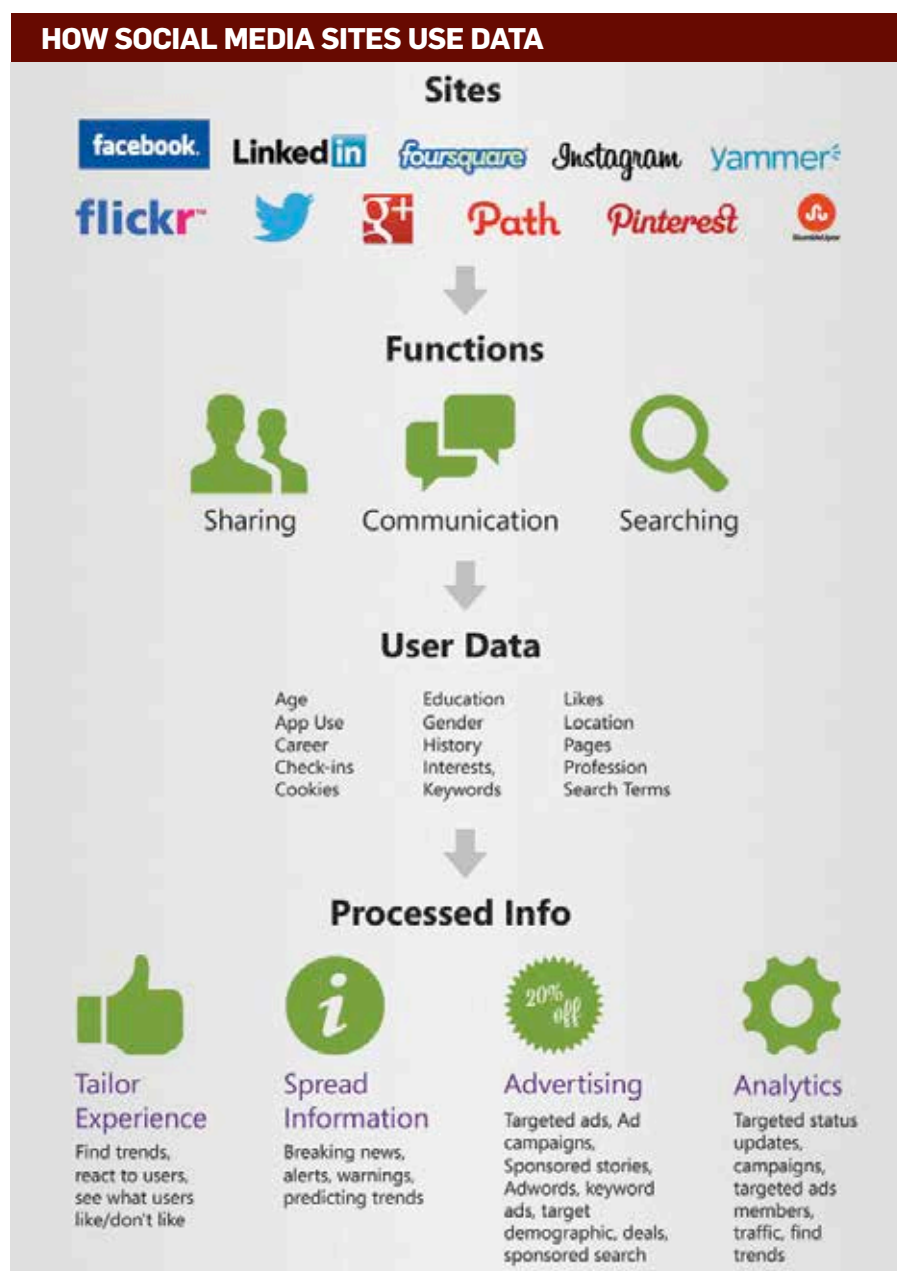
data about yourself is the norm, as privacy is all but a dead concept. When you can't meet friends or family regularly, how do you stay in touch and share your life moments with them if not through popular social networks such as Facebook, Twitter and Google+? We started becoming less careful about what we posted about ourselves online out of a shared convenience experienced by all of us and a misplaced sense of trust with big corporations, who were ultimately controlling and saving all the data we were creating and sharing on their online properties.

This trend birthed the concept of Big Data. IBM likes to define it the following way: "Every day, we create 2.5 quintillion bytes of data – so much that 90 per cent of the data in the world today has been created in the last two years alone. This data comes from everywhere: sensors used to gather climate information, posts to social media sites, digital pictures and videos, purchase transaction records and cell phone GPS signals to name a few. All of this data is essentially Big Data."

As you can imagine, there's plenty of focus on business initiatives and technology practices in companies of all shapes and sizes on Big Data. Simply because of the missing insights it holds and the treasure-trove of information it can provide to add to a company's competitive edge in an increasingly global marketplace. According to a study conducted by Tata Consultancy Services Ltd, two inescapable trends in Big Data provide guidelines for the overall types of data that companies should be looking to gather.

The first trend is internal digital data. This type of data collected from sensors and other remote devices attached to products – a GE aircraft engine or a Xerox copier, for example – enables companies to track their products long after they've been delivered to customers. This provides a major opportunity for companies that can collect unstructured sensor and other data from their products.

Take the classic case of the aviation industry, for instance. An estimated \$284 billion is wasted annually in Aviation by inefficient fuel management, unscheduled aircraft maintenance, flight delays and other issues. Imagine how a company like GE (whose engines are bolted onto 53 per cent of the world's wide-bodied planes)



that can help customers cut their fuel and maintenance costs will no doubt boost market share and increase revenue.

The second trend is data that exists outside the information systems of a company. It's in the hands of customers, third-party data providers, suppliers, social media sites and other sources. To get a holistic picture about their customers, companies must collect this external data. Imagine a chain retailer that can obtain real-time data on customers who are on the move (on their smartphones), within five miles of its stores (the telcos have this data) and about to make a major purchase (such as replacing a refrigerator – the records of

which are owned by the manufacturer). The buyers' decisions are most influenced by reviews in Consumer Reports (which are, of course, found in Consumer Reports' online archives). For that chain retailer to get the customer to visit its store that's nearest to that customer and make a \$2,000 purchase, it needs to tap external data held by the telco, the appliance maker, and Consumer Reports – and in minutes or perhaps even seconds before that customer moves to another store. The internal data available with the retailer isn't nearly enough to influence the consumer's purchasing decision.

Many great insights to be derived from Big Data are likely to come from